Quantum Optical Mirror Memory Simulator

Photon Survival, Squeezing, and Entanglement Preservation

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Purpose: A traceable JupyterLab educational module connecting multipass free-space optical storage to three input quantum resources: a single-photon Fock state, a squeezed vacuum, and a two-mode squeezed vacuum. Physics modules have analytical unit tests; a unified interface explores geometry, loss, and quantum-state quality.

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Abstract

A quantum optical memory must preserve useful quantum-state properties, not merely delay a light pulse. This simulator isolates the minimal quantum physics of a free-space multipass mirror memory: each reflection has intensity survival R, injection and extraction have efficiencies $\eta_{\mathrm{in}}, \eta_{\mathrm{out}}$, and the stored optical mode acquires a deterministic phase. Students compare photon survival, squeezed-quadrature variance, and bipartite entanglement under one common pure-loss channel. The numerical implementation exposes the density matrix for a single photon and covariance matrices for Gaussian states, allowing every diagnostic to be checked against exact physical limits.

Derived impact statement

Impact hypothesis. A mirror-based quantum memory is useful only if its retrieved state retains the quantum resource required by the next operation. This simulator makes the distinction visible: high single-photon survival does not guarantee preservation of squeezing or entanglement, while geometry and cumulative optical loss set measurable limits on the output state.

Executive decision frame

Storage	How do mirror reflectivity, number of reflections, coupling losses, and effective optical path length determine the storage duration and photon retrieval probability?	**Quality**	How does the same optical memory affect a single-photon Fock state, squeezed vacuum, and two-mode entanglement, and which diagnostics quantify useful output quality?

Applications and educational use

Quantum optics courses	Photonic computing	Quantum networking	Simulator verification
Connect propagation and reflection loss to density matrices, squeezing, Gaussian covariance matrices, and quadrature uncertainty.	Explore how finite storage time and loss degrade nonclassical optical resources that may support feed-forward and temporal-mode processing.	Distinguish photon retrieval efficiency from the preservation of nonlocal quantum correlations when one or both optical modes are stored.	Compare exact single-photon and Gaussian-channel formulas with fixed numerical benchmarks and tests of trace, positivity, purity, and symplectic physicality.

Learning objectives

After this module, a learner should be able to connect time to reflection count and optical path; distinguish conditional and unconditional single-photon fidelity; compute retrieved squeezing and Gaussian purity; distinguish entanglement from state purity; and identify assumptions of the independent-loss model.

Scope boundary

The memory is a *model* of a passive free-space optical buffer, not a device-specific simulation of Photon Queue or a full resonant cavity. Reflection loss is independent and frequency-flat; phase shifts are deterministic; no mode mismatch, phase diffusion, detector imperfections, or non-Gaussian dynamics are included. The optical distance between reflections is distinct from physical instrument footprint and from the total unfolded optical path.

1. Physical picture: write, store, read

An optical input mode is injected, travels between reflecting surfaces, and is switched out after a selected propagation duration. Unlike a matter-based memory, the photonic excitation remains optical during storage. If the effective free-space propagation length between consecutive reflections is L and there are N such intervals, the model defines

$$t_N = \frac{NL}{c}, \qquad \eta_N = \eta_{\mathrm{in}}\eta_{\mathrm{out}}R^N, \qquad \phi_N = N\phi_{\mathrm{ref}}. \tag{1}$$

Here R is *intensity* reflectivity, not amplitude reflectivity. The associated surviving field amplitude is $\sqrt{\eta_N}\exp(i\phi_N)$. The idealized input–output relation is

$$\hat{a}_{\mathrm{out}} = \sqrt{\eta_N}\, e^{i\phi_N}\hat{a}_{\mathrm{in}} + \sqrt{1-\eta_N}\,\hat{v}, \tag{2}$$

where $\hat{v}$ is a vacuum environmental mode (with an unimportant choice of environmental phase). This is a bosonic *pure-loss* channel, preserving $[\hat{a}_{\mathrm{out}}, \hat{a}_{\mathrm{out}}^\dagger] = 1$ [1, 2].

Geometry at fixed requested storage time. In the UI, choosing a requested time t produces $N = \lceil ct/L \rceil$ and hence an actual time $t_N \geq t$. At fixed t, increasing L generally lowers N and reflection loss in this model. The result describes *free-space mirror storage*: it does not establish an equivalent scaling law for waveguide delay lines, whose principal loss is often distributed propagation loss. Additional switching, diffraction, alignment, and geometry constraints are omitted.

2. Stage I: a single photon

For an input $|1\rangle$, tracing the inaccessible loss port yields

$$\rho_{\mathrm{out}} = (1-\eta_N)|0\rangle\langle 0| + \eta_N|1\rangle\langle 1|. \tag{3}$$

The unconditional fidelity with the input is $F_{\mathrm{unc}} = \eta_N$, while the conditional fidelity *given successful retrieval* is $F_{\mathrm{cond}} = 1$ for a perfect single-mode pure-loss channel (undefined if $\eta_N = 0$). The output purity is

$$\mu_1 = \mathrm{Tr}\,\rho_{\mathrm{out}}^2 = \eta_N^2 + (1-\eta_N)^2. \tag{4}$$

Although the incident state is pure, loss produces an unheralded mixture of vacuum and one photon. A deterministic phase is globally unobservable for an isolated definite-number state. Equation (3) is also a useful test of trace preservation and positivity.

3. Stage II: squeezed vacuum

Define $\hat{q} = (\hat{a} + \hat{a}^\dagger)/\sqrt{2}$ and $\hat{p} = (\hat{a} - \hat{a}^\dagger)/(i\sqrt{2})$, with vacuum variance $1/2$. An ideal q-squeezed vacuum with real squeezing parameter $r \geq 0$ has covariance

$$V_{\mathrm{in}} = \frac{1}{2}\,\mathrm{diag}(e^{-2r}, e^{2r}). \tag{5}$$

A phase shift rotates quadratures through $\mathcal{R}(\phi)$, giving

$$V_{\text{out}} = \eta_N \mathcal{R}(\phi_N) V_{\text{in}} \mathcal{R}^T(\phi_N) + \frac{1-\eta_N}{2} I_2. \tag{6}$$

The eigenvalues of V are the minimum and maximum quadrature variances. Output squeezing and Gaussian purity are

$$S_{\text{dB}} = -10 \log_{10}(2\lambda_{\min}), \qquad \mu_G = \frac{1}{2\sqrt{\det V}}. \tag{7}$$

For $\phi_N = 0$, $2\lambda_{\min} = 1 - \eta_N + \eta_N e^{-2r}$. An ideal deterministic phase rotation changes the ellipse orientation, not its eigenvalues or purity. A phase-*noise* model would require averaging over random rotations and is not included.

4. Stage III: two-mode entanglement

In quadrature ordering (q_1, p_1, q_2, p_2), the two-mode squeezed vacuum (TMSV) covariance is

$$V_{\text{TMSV}} = \frac{1}{2} \begin{pmatrix} \cosh(2r) I_2 & \sinh(2r) Z \\ \sinh(2r) Z & \cosh(2r) I_2 \end{pmatrix}, \quad Z = \text{diag}(1, -1). \tag{8}$$

Apply the memory independently to mode 1, both modes, or neither. The covariance evolution is $V' = XVX^T + Y$, where each stored mode has block $X_j = \sqrt{\eta_N}\mathcal{R}(\phi_N)$ and $Y_j = (1-\eta_N)I_2/2$; unstored modes use $X_j = I_2$, $Y_j = 0$.

Entanglement is characterized by the smallest symplectic eigenvalue $\tilde{\nu}_-$ of the partially transposed covariance $V^\Gamma = \Lambda V' \Lambda$, $\Lambda = \text{diag}(1, 1, 1, -1)$:

$$E_{\mathcal{N}} = \max\{0, -\log_2(2\tilde{\nu}_-)\}. \tag{9}$$

The state is entangled if $\tilde{\nu}_- < 1/2$ [3, 4]. Other diagnostics include $\mu_2 = 1/\sqrt{\det(2V')}$ and $\Delta^2(q_1 - q_2) + \Delta^2(p_1 + p_2)$. The latter assumes aligned quadrature references and changes under local deterministic phase rotations even when entanglement remains unchanged.

5. Verification and reference benchmark

Reproduce exactly: $R = 0.9999$, $\eta_{\text{in}} = \eta_{\text{out}} = 0.99$, $L = 1.0$ m, $\phi_{\text{ref}} = 0$, $N = 1000$. For the UI's *time* control, use a requested time that returns $N = 1000$, or call the physics methods with $N = 1000$ directly; entering $3.0\,\mu$s yields $N = 900$, not 1000. For squeezed vacuum use 10 dB input ($r = \ln 10/2$). For TMSV use $r = 0.8$ (6.9487 dB on the corresponding single-mode squeezing scale).

Observable	Expected numerical value
Storage time (μs)	3.335641
Total transmissivity	0.886826719
Fock: $P(1)$, $P(0)$	0.886826719, 0.113173281
Fock: unconditional / conditional fidelity	0.886826719 / 1.000000
Fock: purity	0.799269821
Squeezed: minimum variance	0.100928
Squeezed: maximum variance	4.490720
Squeezed: output squeezing (dB)	6.949584
Squeezed: output purity	0.742688
TMSV: no storage — $E_{\mathcal{N}}$, purity	2.308312, 1.000000
TMSV: one mode — $E_{\mathcal{N}}$, purity	2.003967, 0.848517
TMSV: both modes — $E_{\mathcal{N}}$, purity	1.774871, 0.759506
TMSV: EPR variance sums (none / one / both)	0.403793 / 0.502187 / 0.584441

Numerical verification gates: (i) $\mathrm{Tr}\,\rho = 1$, $\rho \succeq 0$; (ii) $\det V \geq 1/4$; (iii) two-mode symplectic eigenvalues $\nu_j \geq 1/2$; (iv) $E_{\mathcal{N}} = 2r/\ln 2$ for lossless TMSV; (v) vacuum remains vacuum for any pure-loss channel. Automated notebook unit tests for all four physical layers and integrated calculation passed in the user-run JupyterLab session. The table above has additionally been recalculated independently from its formulas, to machine precision, not merely transcribed from displayed results.

6. JupyterLab operation and interpretation

Run notebook cells sequentially: (1) `MirrorMemory` class and its tests; (2) Fock density matrix/tests; (3) single-mode Gaussian covariance/tests; (4) two-mode Gaussian covariance, partial transpose and tests; (5) plotting functions; (6A) requested-time-to-reflection helper/tests; (7E) final clean widget UI. An exploratory geometry widget (6B) is optional. The final UI uses fresh widget instances and one update callback, avoiding the earlier controls-display problem.

The dropdown selects Fock, squeezed vacuum, or TMSV. Sliders set R, injection/extraction efficiencies, L, storage time, phase per reflection, and input squeezing; TMSV adds a one/both-mode choice. The UI displays actual reflection count, attained storage duration and relevant quantum diagnostics with plots. **Interpretation note:** it implements control by requested time; any screenshot at a requested $3.0\,\mu$s should show approximately 900 reflections for $L = 1$ m.

7. Educational exercises (with expected outcomes)

Exercise 1: photon survival vs. fidelity. Set $\eta_{\text{in}} = \eta_{\text{out}} = 1$, $R = 0.9999$, $L = 1$ m and $N = 1000$ (or requested $t \approx 3.3356\,\mu$s, checking the rounded count). Predict $P(1)$ and conditional fidelity before running. *Expected:* $P(1) = 0.9999^{1000} = 0.904833$, $P(0) = 0.095167$, and conditional fidelity $= 1$. The mixed output need not have unity purity.

Exercise 2: the geometry tradeoff. Hold requested time $3.0\,\mu$s, $R = 0.9999$, coupling efficiencies 0.99, and vary $L = 0.5, 1, 2$ m. *Expected:* required reflection counts are approximately $1799, 900, 450$; transmissivity increases as L rises (approximately $0.8187, 0.8957, 0.9370$). The improvement follows reduced reflection count; the device footprint is not predicted.

Exercise 3: squeezing is not photon survival. Reproduce the benchmark with 10 dB input squeezing. *Expected:* $\eta = 0.886827$ but only 6.9496 dB retrieved squeezing, with Gaussian purity 0.742688. At $r = 0$,

vacuum always maps to vacuum and purity remains 1.

Exercise 4: deterministic optical phase. For ideal efficiency choose $r = \ln 10/2$, $N = 1$, and vary phase per reflection from 0 to $\pi/2$ using direct functions (the existing UI slider spans only ± 0.1 rad). *Expected:* quadrature axes exchange, while squeezing eigenvalues and purity remain unchanged. The ellipse rotates; no decoherence occurs.

Exercise 5: memory and EPR correlations. With the benchmark and $r = 0.8$, compare no storage, one stored mode, and both stored modes. *Expected:* $E_{\mathcal{N}} = 2.308312, 2.003967, 1.774871$ bits. Entanglement survives despite declining purity. The ideal EPR variance sum is $2e^{-2r} = 0.403793$.

Exercise 6: physicality. Set $R = 0$ and $N \geq 1$. Predict output states before running the Fock, squeezed, and TMSV stages. *Expected:* Fock output is vacuum; squeezed vacuum becomes vacuum; two stored TMSV modes produce independent vacua and zero logarithmic negativity. If only one TMSV mode is lost, the other is thermal, and the bipartite entanglement also vanishes.

8. Model limitations and extensions

The present model assumes a common effective loss coefficient, ideal vacuum loss ports, no memory-induced non-Gaussian transformations, and independent mode losses. The actual free-space geometry may contain distinct reflection angles, switching events, polarization effects, spectral phase shifts and mode-selective coupling. A next version could introduce measured frequency-dependent mirror coefficients $r(\omega)$, phase diffusion, switching loss by circulation, imperfect temporal-mode retrieval, or a comparison with matter-based memory channels. GKP states and genuinely non-Gaussian resources would require more than the Gaussian covariance representation used here.

References

References

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